Tenth Grade Tile Task

Standard G.C.2

10

1. a) Inscribed angle that cuts an arc of 18°: angle? **9**

b) Inscribed angle that cuts an arc of 10°: angle? **5**

c) Inscribed angle that cuts an arc of 4°: angle? **2**

2. a) Inscribed angle that cuts an arc of 2°: angle? **1**

b) Inscribed angle that cuts an arc of 8°: angle? **4**

c) Inscribed angle that cuts an arc of 16°: angle? **8**

3. a) Inscribed angle that cuts an arc of 12°: angle? **6**

b) Inscribed angle that cuts an arc of 14°: angle? **7**

c) Central angle 0°: half of it is the inscribed angle? **0**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Tenth Grade Tile Task

Standard G.C.5

10

1. **a)** Circle circumference 6, arc of 1/1 of the circle: length? **6**

   b) Circle circumference 7, arc of 1/1 of the circle: length? **7**

   c) Circle circumference 4, arc of 1/1 of the circle: length? **4**

2. **a)** Circle circumference 9, arc of 1/1 of the circle: length? **9**

   b) Circle circumference 5, arc of 1/5 of the circle: length? **1**

   c) Circle circumference 4, arc of 1/2 of the circle: length? **2**

3. **a)** Circle circumference 3, arc of 1/1 of the circle: length? **3**

   b) Circle circumference 8, arc of 1/1 of the circle: length? **8**

   c) Circle circumference 5, arc of 1/1 of the circle: length? **5**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Tenth Grade Tile Task

Standard G.CO.1

10

1. **a)** Sides on a pentagon? 5

   b) Sides on a nonagon? 9

   c) Sides on a square? 4

2. **a)** Sides of a ray? 1

   b) Sides on a triangle? 3

   c) Points that define a line? 2

3. **a)** Sides of a line? 0

   b) Sides on a hexagon? 6

   c) Sides on an octagon? 8

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Tenth Grade Tile Task

Standard G.CO.6

10

1. a) Rotational symmetries of a regular hexagon? **6**

b) Rotational symmetries of a regular pentagon? **5**

c) Rotational symmetries of an equilateral triangle? **3**

2. a) Rotational symmetries of a regular octagon? **8**

b) Rotational symmetries of a scalene triangle? **1**

c) Rotational symmetries of a rectangle? **2**

3. a) Rotational symmetries of a square? **4**

b) Rotational symmetries of a regular nonagon? **9**

c) Rotational symmetries of a regular heptagon? **7**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Tenth Grade Tile Task

Standard G.CO.7

10

1. a) Lines of symmetry of a scalene triangle? **0**

b) Sides on a nonagon? **9**

c) Sides on a quadrilateral? **4**

2. a) Angles that must match for AA similarity? **2**

b) Sides that must match for SSS? **3**

c) Sides on a hexagon? **6**

3. a) Right angles in a right triangle? **1**

b) Sides on an octagon? **8**

c) Congruence criteria in the list SSS, SAS, ASA, AAS, HL? **5**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Tenth Grade Tile Task

Standard G.CO.8

10

1. a) Missing angle of a triangle with angles 91° and 86° — **3**

b) Missing angle of a triangle with angles 91° and 82° — **7**

c) Missing angle of a triangle with angles 92° and 86° — **2**

2. a) Missing angle of a triangle with angles 86° and 85° — **9**

b) Missing angle of a triangle with angles 81° and 93° — **6**

c) Missing angle of a triangle with angles 91° and 88° — **1**

3. a) Missing angle of a triangle with angles 82° and 94° — **4**

b) Missing angle of a triangle with angles 94° and 81° — **5**

c) Missing angle of a triangle with angles 81° and 91° — **8**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Tenth Grade Tile Task

Standard G.CO.9

10

1. a) Supplementary angle of 176° **4**

b) Supplementary angle of 177° **3**

c) Vertical angle to a 2° angle (degrees) **2**

2. a) Vertical angle to a 6° angle (degrees) **6**

b) Vertical angle to a 8° angle (degrees) **8**

c) Supplementary angle of 175° **5**

3. a) Supplementary angle of 179° **1**

b) Supplementary angle of 171° **9**

c) Supplementary angle of 173° **7**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Tenth Grade Tile Task

Standard G.CO.10

10

1. a) Triangle angles 86° and 89°: third angle? **5**

b) Triangle angles 88° and 86°: third angle? **6**

c) Triangle angles 85° and 87°: third angle? **8**

2. a) Triangle angles 90° and 81°: third angle? **9**

b) Triangle angles 92° and 85°: third angle? **3**

c) Triangle angles 91° and 82°: third angle? **7**

3. a) Triangle angles 96° and 80°: third angle? **4**

b) Triangle angles 93° and 87°: third angle? **0**

c) Triangle angles 93° and 85°: third angle? **2**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Tenth Grade Tile Task

Standard G.CO.11

10

1. **a)** Opposite angle of a parallelogram with a 6° angle? **6**

   b) Opposite angle of a parallelogram with a 7° angle? **7**

   c) Opposite angle of a parallelogram with a 4° angle? **4**

2. **a)** Opposite angle of a parallelogram with a 2° angle? **2**

   b) Opposite angle of a parallelogram with a 8° angle? **8**

   c) Opposite angle of a parallelogram with a 1° angle? **1**

3. **a)** Opposite angle of a parallelogram with a 9° angle? **9**

   b) Opposite angle of a parallelogram with a 3° angle? **3**

   c) Opposite angle of a parallelogram with a 5° angle? **5**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Tenth Grade Tile Task

Standard G.GMD.1

10

1. a) Diameter of a circle with radius 2? **4**

b) Area of a square of side 1? **1**

c) Radius of a circle with diameter 14? **7**

2. a) Volume of a cylinder, r=1, height 1 (π≈3)? **3**

b) Volume of a cube with side 2? **8**

c) Circumference with diameter 2 (π≈3)? **6**

3. a) Circumference with diameter 3 (π≈3)? **9**

b) Diameter of a circle with radius 1? **2**

c) Radius of a circle with diameter 10? **5**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Tenth Grade Tile Task

Standard G.GMD.3

10

1. a) Volume of a prism 3 by 3 by 1 — 9

b) Volume of a prism 1 by 1 by 1 — 1

c) Volume of a prism 5 by 1 by 1 — 5

2. a) Volume of a prism 2 by 2 by 2 — 8

b) Volume of a prism 7 by 1 by 1 — 7

c) Volume of a prism 1 by 1 by 4 — 4

3. a) Volume of a prism 3 by 1 by 1 — 3

b) Volume of a prism 1 by 1 by 2 — 2

c) Volume of a prism 1 by 6 by 1 — 6

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Tenth Grade Tile Task

Standard G.GPE.4

10

1. **a)** x-coordinate of point (9, 1) — 9

   b) x-coordinate of point (5, 2) — 5

   c) x-coordinate of point (6, 1) — 6

2. **a)** x-coordinate of point (2, 0) — 2

   b) x-coordinate of point (7, 0) — 7

   c) x-coordinate of point (0, 1) — 0

3. **a)** x-coordinate of point (8, 1) — 8

   b) x-coordinate of point (1, 0) — 1

   c) x-coordinate of point (4, 0) — 4

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Tenth Grade Tile Task

Standard G.GPE.6

10

1. **a)** Midpoint of (0, 0) and (0, 0): y? **0**

   b) Midpoint of (0, 12) and (0, 0): y? **6**

   c) Midpoint of (0, 1) and (0, 13): y? **7**

2. **a)** Midpoint of (0, 2) and (0, 4): y? **3**

   b) Midpoint of (0, 4) and (0, 0): y? **2**

   c) Midpoint of (0, 7) and (0, 3): y? **5**

3. **a)** Midpoint of (0, 7) and (0, 11): y? **9**

   b) Midpoint of (0, 0) and (0, 2): y? **1**

   c) Midpoint of (0, 16) and (0, 0): y? **8**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Tenth Grade Tile Task

Standard G.GPE.7

10

1. **a)** Distance from (0, 0) to (0, 0) — 0

   b) Distance from (0, 0) to (0, 8) — 8

   c) Distance from (0, 0) to (0, 5) — 5

2. **a)** Distance from (0, 0) to (0, 1) — 1

   b) Distance from (0, 0) to (0, 2) — 2

   c) Distance from (0, 0) to (0, 7) — 7

3. **a)** Distance from (0, 0) to (0, 4) — 4

   b) Distance from (0, 0) to (0, 6) — 6

   c) Distance from (0, 0) to (0, 9) — 9

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Tenth Grade Tile Task

Standard G.MG.1

10

1. **a)** Sides of a regular 3-gon used as a model? **3**

   b) Curved sides on a model of a circle? **1**

   c) Straight sides on a model of a circle? **0**

2. **a)** Sides of a regular 9-gon used as a model? **9**

   b) Sides of a regular 4-gon used as a model? **4**

   c) Sides of a regular 6-gon used as a model? **6**

3. **a)** Sides of a regular 7-gon used as a model? **7**

   b) Sides of a regular 5-gon used as a model? **5**

   c) Parallel sides on a model of a rectangle? **2**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Tenth Grade Tile Task

Standard G.SRT.2

10

1. **a)** Scale factor from a side of 7 to a side of 14 — 2
   b) Scale factor from a side of 2 to a side of 14 — 7
   c) Scale factor from a side of 3 to a side of 9 — 3

2. **a)** Scale factor from a side of 4 to a side of 16 — 4
   b) Scale factor from a side of 2 to a side of 18 — 9
   c) Scale factor from a side of 3 to a side of 18 — 6

3. **a)** Scale factor from a side of 1 to a side of 8 — 8
   b) Scale factor from a side of 4 to a side of 4 — 1
   c) Scale factor from a side of 3 to a side of 15 — 5

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Tenth Grade Tile Task

Standard G.SRT.4

10

1. a) Midsegment of a triangle with base 4? **2**

b) Midsegment of a triangle with base 12? **6**

c) Midsegment of a triangle with base 18? **9**

2. a) Hypotenuse of a 3-4-5 triangle? **5**

b) Midsegment of a triangle with base 14? **7**

c) Midsegment of a triangle with base 8? **4**

3. a) Midsegment of a triangle with base 2? **1**

b) Midsegment of a triangle with base 6? **3**

c) Midsegment of a triangle with base 16? **8**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Tenth Grade Tile Task

Standard G.SRT.5

10

1. a) Missing angle of a triangle with angles 84° and 92° — **4**

b) Missing angle of a triangle with angles 93° and 85° — **2**

c) Missing angle of a triangle with angles 90° and 81° — **9**

2. a) Missing angle of a triangle with angles 81° and 99° — **0**

b) Missing angle of a triangle with angles 85° and 88° — **7**

c) Missing angle of a triangle with angles 86° and 93° — **1**

3. a) Missing angle of a triangle with angles 90° and 82° — **8**

b) Missing angle of a triangle with angles 95° and 80° — **5**

c) Missing angle of a triangle with angles 81° and 96° — **3**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Tenth Grade Tile Task

Standard G.SRT.6

10

1. a) In a 6-8-10 triangle, opposite leg? **6**

b) Tangent of 45°: numerator when legs are 2 and 2? **2**

c) Tangent of 45°: numerator when legs are 1 and 1? **1**

2. a) Tangent of 45°: numerator when legs are 7 and 7? **7**

b) In a 9-12-15 triangle, opposite leg? **9**

c) In a 6-8-10 triangle, adjacent leg? **8**

3. a) In a 3-4-5 triangle, cos numerator (adjacent)? **4**

b) In a 3-4-5 triangle, sin numerator (opposite)? **3**

c) In a 3-4-5 triangle, hypotenuse? **5**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Tenth Grade Tile Task

Standard G.SRT.8

10

1. a) Hypotenuse 5, leg 4: other leg? **3**

b) Hypotenuse 10, leg 6: other leg? **8**

c) Area of a square with side 1? **1**

2. a) Area of a right triangle with legs 5 and 2? **5**

b) Hypotenuse 10, leg 8: other leg? **6**

c) Area of a right triangle with legs 7 and 2? **7**

3. a) Area of a right triangle with legs 2 and 4? **4**

b) Area of a right triangle with legs 2 and 2? **2**

c) Area of a square with side 3? **9**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9